

Spring 2026 Natural Language Processing (CSCI-6515)

Project 3: Word Embeddings, Comparative Analysis, DL Classification, and Interactive UI

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Outline

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- Azerbaijani NLP needs stronger semantic representations than sparse token counts.
- We evaluate embeddings at two levels:
 - Intrinsic: nearest neighbors + analogy arithmetic.
 - Extrinsic: downstream sentiment classification.
- Project 3 objective: complete Tasks 1–6 and provide an interactive UI for exploration.

Data sources

- Corpus for Tasks 1–3: `data/raw/corpus.csv`
- Sentiment for Tasks 4–5: `data/external/train.csv`, `test.csv`

Pipeline orchestration

- Runner: `bash scripts/run_project3.sh`
- Task outputs: `outputs/project3/task1_...task6_...`
- UI: Streamlit dashboard + legacy Flask UI

Task 1 Method (with light math)

Term-document matrix

$$X \in \mathbb{R}^{D \times V}, \quad X_{d,w} = \text{count}(w \text{ in document } d)$$

Word-word co-occurrence matrix

$$C_{i,j} = \sum_t 1[w_t = i]1[w_{t+\Delta} = j], \quad |\Delta| \leq W$$

Configured thresholds

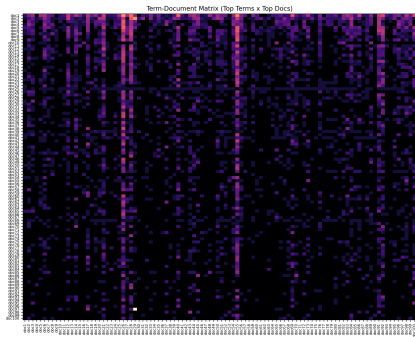
- Frequent words: $\text{count} \geq 100$
- Rare words: $\text{count} \leq 2$
- Heatmap view: top terms/docs for readability

Task 1 Results

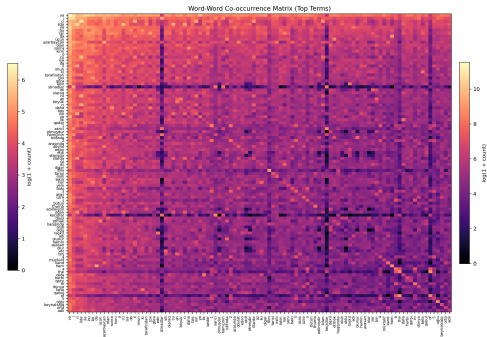
Metric	Value
Documents	31,842
Sentences	813,319
Tokens	11,905,937
Vocabulary size	586,674
Frequent words (≥ 100)	12,436
Rare words (≤ 2)	398,599

Artifacts: term-document sparse matrix, word frequency distribution, top-matrix CSVs, and heatmaps.

Task 1 Visualizations



Term-document heatmap



Word-word heatmap

Task 2 Word2Vec Setup

Training objective (skip-gram)

$$\max_t \sum_{-k \leq j \leq k, j \neq 0} \log p(w_{t+j} \mid w_t)$$

Configuration

- Model: skip-gram (cbow=0)
- Dimension: 200, Window: 5
- Negative sampling: 10, Min count: 5
- Iterations: 10

Evaluation

- Similar words for 10 target words
- Analogy arithmetic ($b - a + c$)

Task 2 Results (Word2Vec)

- Vocab size: 122,930, vector dim: 200
- Analogy hit@k: 0.6667 (3 evaluated / 5 total)
- Good local morphology for several targets (e.g., seher → seheri)
- Some noisy neighbors remain (data noise + OOV effects)

Example analogies

- Correct: kishi:qadin :: oglan:qiz
- Correct: baki:azerbaycan :: ankara:turkiye
- Incorrect: ata:ana :: ogul:? → isterem

Task 3 GloVe Setup

GloVe objective

$$J = \sum_{i,j} f(X_{ij}) \left(w_i^\top \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

Configuration

- Dimension: 200, Window: 10
- Min count: 5, Iterations: 20
- $x_{\max} = 10$, $\eta = 0.05$, $\alpha = 0.75$

Same evaluation protocol as Task 2

Task 3 Results (GloVe)

- Vocab size: 122,930, vector dim: 200
- Analogy hit@k: 0.7500 (4 evaluated / 5 total)
- Strong morphology neighbors also observed for many targets
- Some high-score noisy neighbors still appear

Example analogies

- Correct: kishi:qadin :: oglan:qiz
- Correct: shimal:cenub :: sherq:qerb
- Incorrect: ata:ana :: ogul:? → isterem

Task 4: Word2Vec vs GloVe

Model	Hit@1	Hit@k	Avg top-3 cosine	Coverage
Word2Vec	0.6667	0.6667	0.6619	0.0749
GloVe	0.7500	0.7500	0.6217	0.0749

Interpretation

- GloVe is better on analogy score.
- Word2Vec is better on local-neighbor cosine quality.
- Coverage is equal; downstream choice depends on objective.

Task 5 Experimental Design

Architectures

- RNN, BiRNN, LSTM

Feature families

- Count-SVD, TFIDF-SVD, PMI-SVD, Word2Vec, GloVe

Total runs

- 15 neural runs (3x5) + baseline (LogReg + TFIDF)

Primary metric

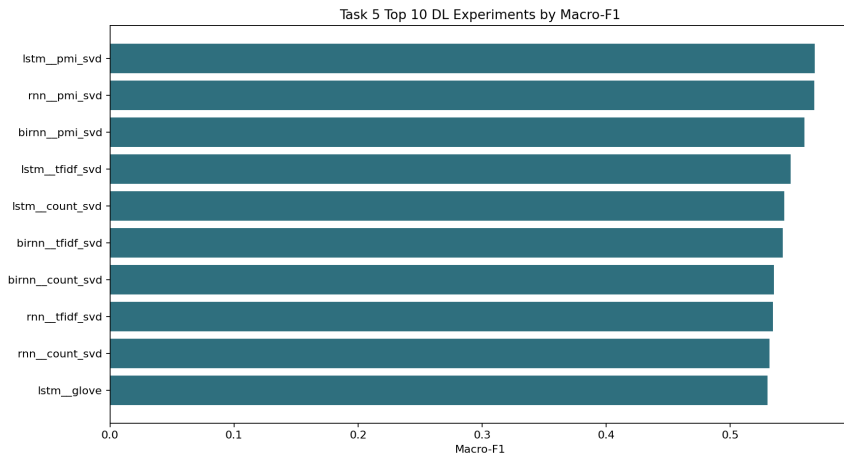
$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c$$

Task 5 Results

Experiment	Accuracy	Macro-F1	Weighted-F1
lstm__pmi_svd	0.8225	0.5684	0.8594
rnn__pmi_svd	0.8396	0.5678	0.8641
birnn__pmi_svd	0.8170	0.5600	0.8541
logreg_baseline__tfidf	0.8271	0.5588	0.8582

Best run: lstm__pmi_svd by macro-F1.

Task 5 Visualization



Task 6 Deliverables

- Consolidated report tables:
 - `embedding_comparison_table.csv`
 - `dl_results_table.csv`
- Report note artifact: `report_artifacts.md`
- Presentation bundle zip with figures/tables/summaries

Extra Task: Interactive UI

Primary UI: Streamlit dashboard (`src/project3_dashboard.py`)

- Overview and task tabs for Task 1–6 outputs
- Embedding playground:
 - custom nearest-neighbor queries
 - custom analogy solver
 - vector blending and semantic map (2D)
- Read-only artifact browsing (no retraining from UI)

Limitations and Improvement Areas

- Sentiment dataset is highly imbalanced (positive dominates).
- Intrinsic analogy set is relatively small and OOV-sensitive.
- Some nearest-neighbor outputs are noisy due corpus artifacts.
- Next steps:
 - curated Azerbaijani analogy benchmark
 - better class balancing / reweighting
 - transformer baselines for Task 5

- All Project 3 tasks were implemented with reproducible outputs.
- Word2Vec and GloVe both learned useful structure with different strengths.
- Best downstream model: `lstm_pmi_svd` ($F1_{\text{macro}} = 0.5684$).
- UI enables interactive model/result exploration for presentation and grading.

Thank you!

Questions?